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A highly selective fluorescent chemosensor based on naphthalimide and Schiff base units for Cu²⁺ detection in aqueous medium

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Abstract: A new Schiff base, 4-allylamine-*N*-(*N*-5-methylsalicylidene)-1,8-naphthalimide (**1**), has been designed and synthesized by combining a 1,8-naphthalimide moiety as a fluorophore and a Schiff base as a recognition group. Its photophysical properties were investigated by absorption and fluorescence spectroscopy, and this sensor exhibits a high fluorescence quantum yield of 0.75–0.91 inorganic solvents of different polarity. It also shows high selectivity for Cu²⁺ over other ions with fluorescence quenching in aqueous medium (pH = 7.2). The reason for this phenomenon (fluorescence quenching) is attributed to the formation of a 1 : 1 complex between **1** and Cu²⁺ according to the Job plot and fluorescence titration. The sensor can be applied to the quantification of Cu²⁺ in a linear fashion from 0.5 to 5 μM with a detection limit of 0.23 μM. Additionally, the association constant (K_a) between Cu²⁺ and **1** is $1.328 \times 10^6 \text{ M}^{-1}$ in aqueous media.

Keywords: chemosensor; Cu²⁺ ion; fluorescence; naphthalimide; Schiff base.

1 Introduction

As the third most abundant essential trace element after iron and zinc in the human body, copper performs an important role in many fundamental physiological processes in the human body, such as functional and structural strengthening of protein and gene expression, even in the human body's nervous system [1, 2]. It is well known that it serves as a catalytic cofactor for a variety

of metalloenzymes [3], but unregulated large intake of copper can result in severe neurodegenerative diseases such as Menkes and Wilson diseases, familial amyotrophic lateral sclerosis, Alzheimer's disease and prion diseases, and can cause infant liver damage [4–6]. Thus, the selective measurement and monitoring of Cu²⁺ in environmental and biological samples are of considerable significance for environment protection and human health.

Till date, the available methods for Cu²⁺ detection include colorimetry [7–12], electrochemistry [13, 14], room temperature phosphorimetry [15], atomic absorption spectrometry [16, 17], inductive coupled plasma-mass spectrometry [18] and fluorescence sensing [19–23]. During the past few decades, the fluorescence chemosensors, which combine a recognition functional moiety with optical transduction, have received much attention as an efficient analytical technique for the detection of a particular species due to their high selectivity and sensitivity, intrinsic specificity, real-time monitoring and fast response time [24–26]. Thereby, many fluorescence sensors based on quinoline [27], fluorescein [28], pyrene [29, 30], rhodamine [31, 32], azobenzene [33], coumarin [34–36], 1,8-naphthalimide [37–39], anthraquinone [40, 41] and other fluorophores [42, 43] for Cu²⁺ sensing have been synthesized and investigated. However, there is still an intense demand for efficient Cu²⁺ fluorescence chemosensors, especially those that can work in aqueous solution with high selectivity and sensitivity.

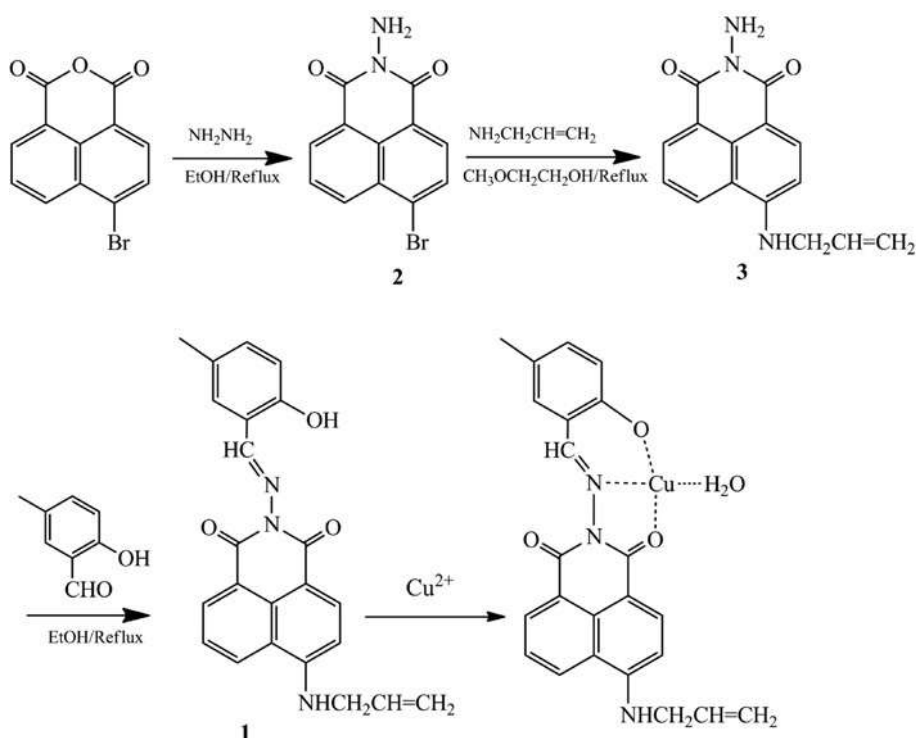
Herein, we developed a new sensor based on a naphthalimide framework. The selectivity of **1** towards various transition metal cations was investigated. It could be used to detect and recognize a Cu²⁺ ion with high selectivity and a low detection limit in aqueous solution.

2 Results and discussions

The synthesis of compound **1** was achieved in three steps as shown in Scheme 1. It is soluble in dipolar aprotic solvents such as THF, DMF, DMSO, acetonitrile, acetone and dichloromethane, and insoluble in Et₂O and water. The

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Scheme 1: Chemical structure of and synthetic route to compound **1**.

structure and purity of the synthesized compound **1** were confirmed by conventional techniques – melting point, thin-layer chromatography (TLC) (R_f values) and UV/Vis spectra – and identified by ^1H and ^{13}C NMR spectra, FT-IR and elemental analysis data. The data are presented in Section 3.

2.1 Photophysical characteristics of **1**

The photophysical properties of the substituted 1,8-naphthalimides are known to depend mainly on the polarization of their chromophoric system. Light absorption in this molecule generates a charge transfer interaction between the substituents at C-4 position and the imide carbonyl groups. Therefore, the photophysical characteristics of **1** in organic solvents with different polarity have

been investigated. Table 1 exhibits its absorption (λ_A) and fluorescence (λ_F) maxima, the extinction coefficient (ϵ), the Stokes shift ($\nu_A - \nu_F$) and the quantum fluorescence yield (Φ_F).

Compound **1** absorbs in the near-UV region with maxima at $\lambda_A = 426\text{--}440\text{ nm}$, and the absorption can be ascribed to $\pi\text{--}\pi^*$ transitions of the naphthalene ring. The respective fluorescence maxima are at $502\text{--}526\text{ nm}$. The molar extinction coefficient (ϵ) of $8422\text{--}18\,603\text{ M}^{-1}\text{ cm}^{-1}$ corresponds to an $S_0 \rightarrow S_1$ transition.

The polarity of the organic solvents is of great importance for the photophysical properties of **1** under study and especially for the quantum fluorescence yield and Stokes shift. The Stokes shift is an important parameter, which indicates the difference in the properties and structure of the fluorophore between the ground state S_0 and the first excited state S_1 . The Stokes shift was calculated by eq. 1.

$$(\nu_A - \nu_F) = (1/\lambda_A - 1/\lambda_F) \times 10^7 \text{ cm}^{-1} \quad (1)$$

Table 1: Photophysical properties of **1** in organic solvents with different polarity.

Organic solution	λ_A (nm)	ϵ ($\text{M}^{-1}\text{ cm}^{-1}$)	λ_F (nm)	$\nu_A - \nu_F$ (cm^{-1})	Φ_F
THF	429	13 263	508	3625	0.91
DMF	440	18 603	526	3716	0.75
Acetonitrile	433	11 751	522	3938	0.76
Acetone	432	8422	517	3806	0.80
Dichloromethane	426	11 223	502	3554	0.77

The obtained Stokes shift values are in the $3554\text{--}3938\text{ cm}^{-1}$ region. Table 1 reveals that the Stokes shift depends on the media and it is larger in the case of polar solvents where hydrogen bond formation or dipole–dipole interactions are favored in comparison to non-polar media. The ability of **1** to emit absorbed light energy is characterized quantitatively by the quantum yield of fluorescence

Φ_F . The fluorescence quantum yield has been calculated on the basis of the absorption and fluorescence spectra using *N*-butyl-4-*n*-butylamino-naphthalimide ($\Phi_F = 0.81$ in ethanol) according to eq. 2.

$$\Phi_F = \Phi_{ref} (S_{sample} / S_{ref}) (A_{ref} / A_{sample}) (n_{sample} / n_{ref})^2 \quad (2)$$

Herein, Φ_F is the emission quantum yield of the sample, Φ_{ref} is the emission quantum yield of the standard, A_{ref} and A_{sample} represent the absorbance of the standard and sample at the excited wavelength, respectively, whereas S_{ref} and S_{sample} are the integrated emission band areas of the standard and sample, respectively, and n_{ref} and n_{sample} are the solvent refractive indices of the standard and sample, respectively. It is seen that the polarity of the environment has no obvious influence on the Φ_F values (Table 1).

2.2 Influence of pH on the photophysical properties of 1

The spectroscopic characteristics of compound **1** were performed in the pH range 1.81–11.82 in Britton–Robinson buffer–DMF (*v/v* = 1:1) solution. The fluorescence emission spectra were recorded at room temperature with an excitation at 430 nm. It is obvious from Fig. 1 that the fluorescence intensity (539 nm) of **1** decreases dramatically when the pH value goes down from 3.78 to 1.81 or goes up from 7.96 to 11.82. However, the fluorescence intensity of **1** is high and relatively stable in the pH range from 3.78 to 7.96, implying that **1** was actually pH-independent between pH 3.78 and 7.96. Therefore, the following experiments were carried out in solution at pH 7.2.

2.3 The fluorescence emission spectroscopy of 1 to Cu²⁺

High selectivity for a specific analyte in the presence of a competing species is an important feature of a chemosensor. To examine the selectivity of **1** (5 μ M) towards various transition metal ions [Na⁺, K⁺, Ca²⁺, Mg²⁺, Al³⁺, Pb²⁺, Fe³⁺, Ni²⁺, Zn²⁺, Cu²⁺, Hg²⁺, Ag⁺, Co²⁺, Cr³⁺, Mn²⁺ and Cd²⁺ (10 μ M, 2 eq)], the fluorescence spectroscopic properties were investigated in Tris-HCl (pH = 7.2) buffer–DMF (1:1, *v/v*) solution. The fluorescence emission spectra were collected at room temperature with an excitation at 430 nm. Both the excitation and emission slit widths were 1.0 nm. As seen from Fig. 2a, we found that only Cu²⁺ caused a noteworthy fluorescence decrease at 539 nm, whereas other metal ions induced only negligible fluorescence variations.

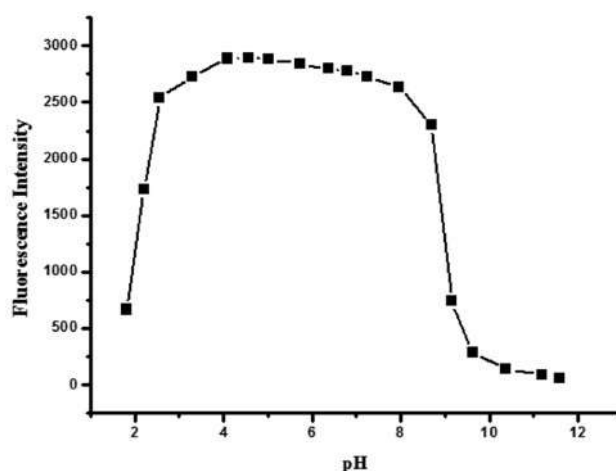


Fig. 1: Photophysical behavior of **1** in Britton–Robinson buffer–DMF (1:1) solution at different pH values at 539 nm.

The selectivity difference can be assigned to the different binding abilities of these metal ions to this sensor. These results indicate that **1** could be used as a Cu²⁺ selective fluorescent sensor.

To further examine the affinity of **1** to Cu²⁺, competitive binding between Cu²⁺ and other metals to **1** was measured by using a mixed solution containing Cu²⁺ (10 μ M) and each of the other metal cations (Na⁺, K⁺, Ca²⁺, Mg²⁺, Al³⁺, Pb²⁺, Fe³⁺, Ni²⁺, Zn²⁺, Hg²⁺, Ag⁺, Co²⁺, Cr³⁺, Mn²⁺ and Cd²⁺) at a concentration of 10 μ M in Tris-HCl (pH = 7.2) buffer–DMF (1:1, *v/v*) solution. From Fig. 2b, we can find that all competitive metal ions had no obvious interference with the detection of a Cu²⁺ ion, which indicated that the compound **1**–Cu²⁺ system was hardly affected by these ions.

In order to get further insight into the binding interaction between **1** and Cu²⁺, a fluorescence titration of **1** with Cu²⁺ was carried out and showed an emission maximum at 539 nm. It can be seen from Fig. 2c that on the addition of a Cu²⁺ ion, the fluorescence intensity of **1** gradually decreased, and when the amount of Cu²⁺ ion added was about 5×10^{-6} M, the fluorescence intensity reached a minimum. When more Cu²⁺ was added, the fluorescence intensity showed negligible changes. The nonlinear curve fitting of the fluorescence titration gives a 1:1 stoichiometric ratio between compound **1** and Cu²⁺. Moreover, a Job's plot [44], which exhibits a maximum at 0.5 M fraction of Cu²⁺, further indicates that only a 1:1 complex is formed (Fig. 2d).

Based on the fluorescence titration of **1** with Cu²⁺, the association constant has been calculated to be 1.328×10^6 M⁻¹ (error limits $\leq 10\%$) by a Benesi–Hildebrand equation [45–47] (Fig. 3a and eq. 3).

$$1/(F - F_0) = 1/\{K_a (F_m - F_0)[Cu^{2+}]^n\} + 1/(F_m - F_0) \quad (3)$$

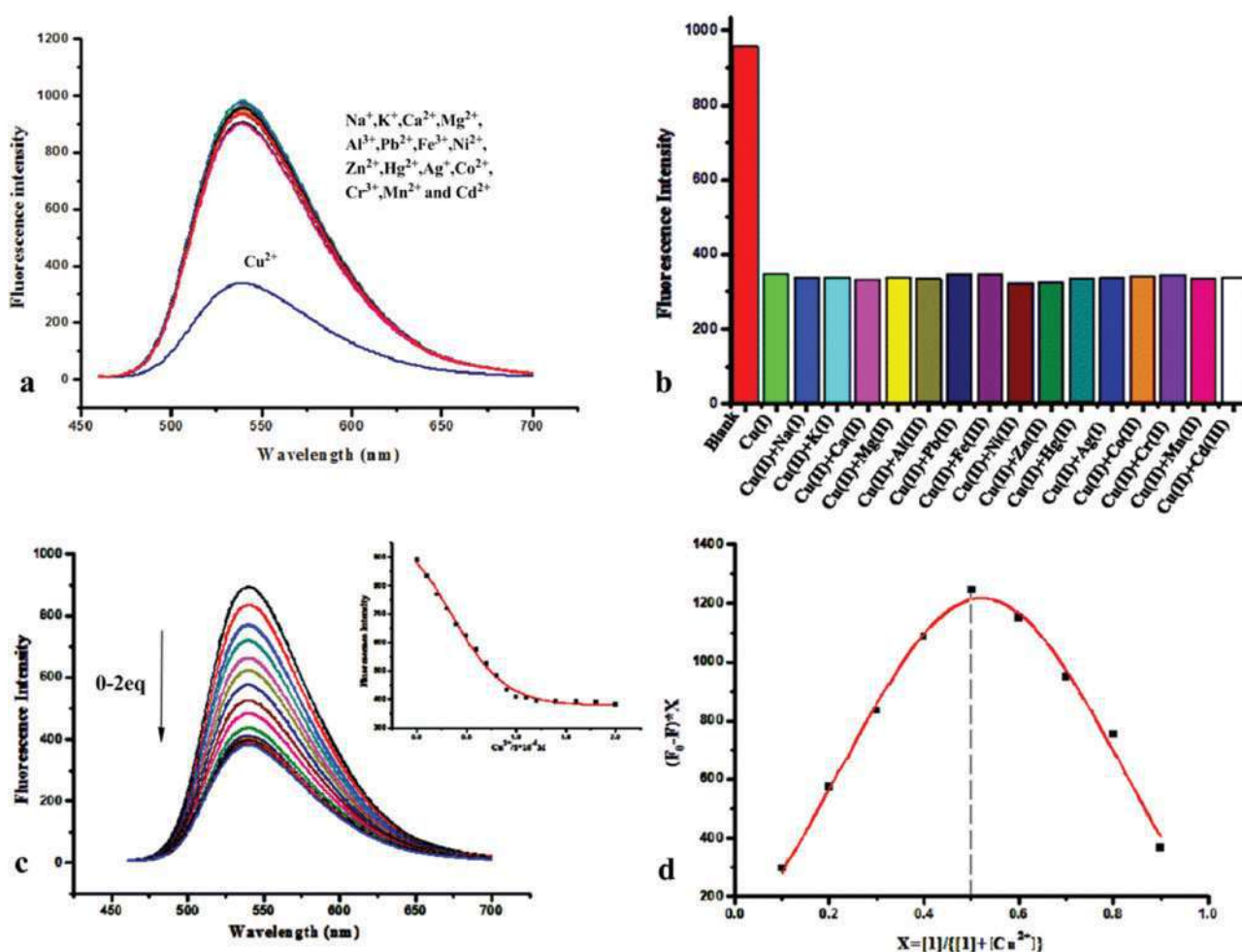


Fig. 2: (a) Fluorescence spectra of **1** (5×10^{-6} M) in Tris-HCl (pH=7.2) buffer-DMF (1:1, v/v) solution towards Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Al^{3+} , Pb^{2+} , Fe^{3+} , Ni^{2+} , Zn^{2+} , Cu^{2+} , Hg^{2+} , Ag^+ , Co^{2+} , Cr^{3+} , Mn^{2+} or Cd^{2+} (2 eq) at 539 nm ($\lambda_{\text{ex}} = 430$ nm); (b) Competitive experiments in the compound **1** + Cu^{2+} system with interfering metal ions at 539 nm ($\lambda_{\text{ex}} = 430$ nm); (c) Fluorescence titration spectra of **1** with Cu^{2+} (0–2 eq) in Tris-HCl (pH=7.2) buffer-DMF (1:1, v/v) solution at 539 nm ($\lambda_{\text{ex}} = 430$ nm); (d) Job's plot at 539 nm ($\lambda_{\text{ex}} = 430$ nm).

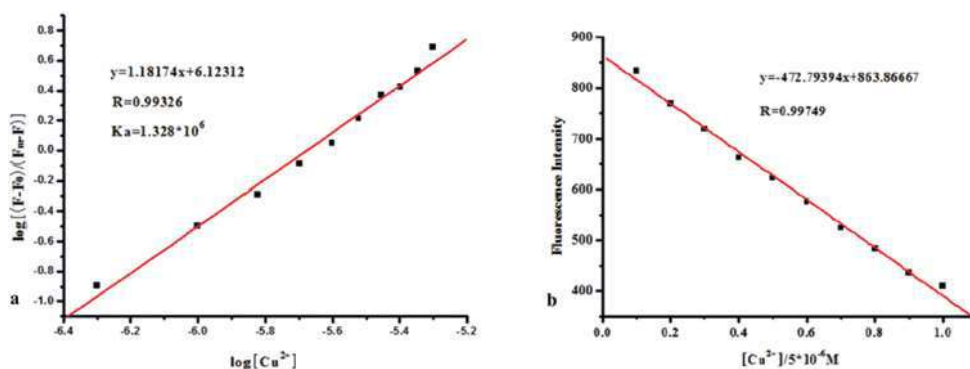


Fig. 3: (a) Benesi-Hildebrand linear analysis plots of **1** at different Cu^{2+} concentrations; (b) curve of fluorescence intensity at 539 nm of **1** (5×10^{-6} M) versus increasing concentrations of Cu^{2+} (0.5–5 μM).

Herein, F is the fluorescence intensity at 539 nm at any given Cu^{2+} concentration, F_0 is the fluorescence intensity at 539 nm in the absence of Cu^{2+} and F_m is the minimum

fluorescence intensity at 539 nm in the presence of Cu^{2+} in solution. The association constant K_a was evaluated graphically by plotting $\log[(F - F_0)/(F_m - F)]$ against $\log[\text{Cu}^{2+}]$.

Figure 3b also shows good linearity relating the emission at 539 nm and the concentrations of Cu²⁺ in the range from 0.5 to 5 μ M, indicating that **1** can detect quantitatively relevant concentrations of Cu²⁺.

The detection limit based on the definition by IUPAC was calculated by the Stern–Volmer plot [48]:

$$C_{DL} = 3\sigma/k = 0.23 \mu\text{M}$$

where σ is the standard deviation of the blank solution, and k is the slope between F_0/F and [Cu²⁺]. In order to determine σ , the emission intensity of **1** in Tris-HCl (pH=7.2) buffer–DMF (1:1, v/v) solution without any metal ions was measured 10 times.

2.4 The UV/Vis absorption spectroscopic response of **1** to Cu²⁺

Additionally, a UV/Vis titration experiment of **1** was performed with Cu²⁺ in Tris-HCl (pH=7.2) buffer–DMF (1:1, v/v) solution. Figure 4 shows the UV/Vis absorption spectra of **1** (5×10^{-6} M) in the presence of various concentrations of Cu²⁺ ion ($0-1 \times 10^{-5}$ M, 0–2 eq), and the inset shows the plots of changes in the 334 and 376 nm maxima as a function of increasing concentrations of Cu²⁺. The absorbance of **1** at 334 nm gradually decreases with an increasing concentration of Cu²⁺ ions; this change is clear evidence of C=N coordination to a Cu²⁺ ion. Owing to the coordination of **1** with a Cu²⁺ ion, the absorbance at 376 nm increases with an

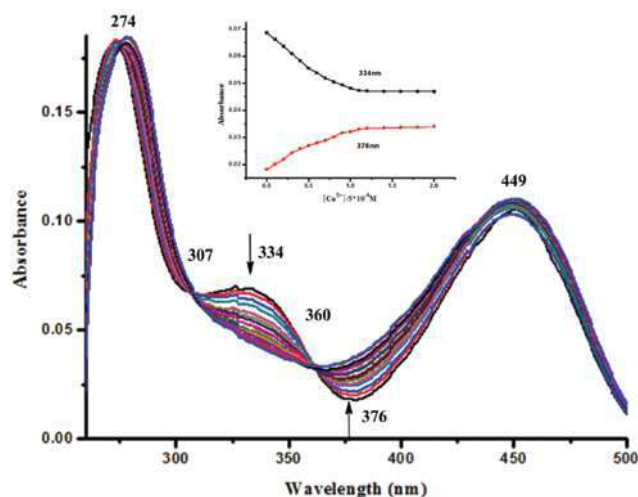


Fig. 4: UV/Vis titration of **1** (5×10^{-6} M) in Tris-HCl (pH=7.2) buffer–DMF (1:1, v/v) solution with increasing amount of Cu²⁺ ($0-1 \times 10^{-5}$ M, 0–2 eq); the absorbance plot of **1** against increasing [Cu²⁺] at $\lambda_{334\text{nm}}$ and $\lambda_{376\text{nm}}$ is shown in the inset.

increasing concentration of Cu²⁺ ions. As Cu²⁺ was gradually titrated, the absorbance of **1** at 334 and 376 nm reached extrema when the amount of Cu²⁺ ion added was about 5×10^{-6} M. When more Cu²⁺ was titrated, the absorbance showed negligible changes, implying that a 1:1 complex was formed. Moreover, two isosbestic points appear at 307 and 360 nm. The absorptions at 274 and 449 nm can be attributed to $\pi-\pi^*$ transitions of the benzene and naphthalene rings, respectively.

Coupling fluorescence titration, Job's plot and UV/Vis titration analysis together, we could primarily confirm that the 1:1 stoichiometry for the **1**–Cu(II) complex was formed as shown in Scheme 1. The results obtained were consistent with the reported reference [49].

3 Experimental section

3.1 Apparatus and reagents

Absorption spectra were scanned using a Lab-Tech UV Bluestar spectrophotometer with one pair of 10-mm quartz cells at room temperature. The IR spectra were recorded in the 4000–400 cm^{-1} region with a Nicolet FT-VERTEX 70 spectrometer using KBr pellets. Mass spectra were recorded on a Mass Spectrometer micrOTOF. ¹H NMR and ¹³C NMR spectra were acquired on a Mercury plus 400 MHz NMR spectrometer with tetramethylsilane (TMS) as an internal standard and [D₆]DMSO as solvent. The corrected excitation and fluorescence spectra were taken on an F97Pro fluorescence spectrophotometer. A quartz cuvette ($1 \times 1 \text{ cm}^2$) was used for the spectroscopic analysis. TLC was performed on silica gel, Fluka F60 254, 20 $\times$ 20, 0.2 mm. The melting points were determined by means of a Kofler melting point microscope.

All chemicals and solvents used in the synthesis were analytical reagent grade and used without further purification. *N*-Butyl-4-*n*-butylamino-naphthalimide ($\Phi_F = 0.81$ in ethanol) was synthesized according to the literature [50]. Britton–Robinson buffer was prepared with 40 mM acetic acid, boric acid and phosphoric acid. Dilute hydrochloric acid or sodium hydroxide was used to adjust pH values. Tris-HCl buffer (pH=7.2) was prepared using bidistilled water. The metal cation sources, NaNO₃, KNO₃, CaCl₂, MgSO₄, AlCl₃, Pb(NO₃)₂, Fe(NO₃)₃·9H₂O, Ni(NO₃)₂·6H₂O, Zn(NO₃)₂·6H₂O, Cu(NO₃)₂·5H₂O, Hg(NO₃)₂·H₂O, AgNO₃, Co(NO₃)₂·6H₂O, Cr(NO₃)₃·9H₂O, 50% Mn(NO₃)₂ and Cd(NO₃)₂·H₂O, were analytical reagent grade and were dissolved in bidistilled water.

3.2 Synthesis

The solution of 4-bromo-1,8-naphthalic anhydride (13.85 g, 50 mmol) and 80% hydrazine hydrate (6.25 g, 100 mmol) in 150 mL of ethanol was refluxed with stirring for 4 h [51]. The reaction's progress was monitored by TLC (with CH₂Cl₂ as the eluent). The mixture was cooled and the precipitated solids were filtered, recrystallized from ethanol and dried to give 14.5 g (85.6%) of *N*-amido-4-bromo-1,8-naphthalimide (**2**) as yellow-brown crystals. – ¹H NMR (CDCl₃): δ (ppm) = 8.68 (d, 1H, *J* = 7.2 Hz), 8.60 (d, 1H, *J* = 8.4 Hz), 8.44 (d, 1H, *J* = 7.6 Hz), 8.06 (d, 1H, *J* = 8.06 Hz), 7.87 (t, 1H, *J* = 7.6 Hz), 5.30 (s, 2H).

N-Amido-4-allylamino-1,8-naphthalimide (**3**) was prepared from 7.35 g (25 mmol) of *N*-amido-4-bromo-1,8-naphthalimide (**2**), 7.13 g (75 mmol) of allylamine and 100 mL of 2-methoxyethanol. The resulting mixture was refluxed and stirred for 64 h and then poured into 500 mL of water. The precipitate was collected by filtration, washed with water and dried to give 6.25 g (93%) of *N*-amido-4-allylamino-1,8-naphthalimide (**3**); m.p.: 180–182°C. – Elemental analysis for C₁₅H₁₃N₃O₂: calcd. C 67.41, H 4.90, N 11.72; found C 67.93, H 5.34, N 12.34.

N-Amido-4-allylamino-1,8-naphthalimide (**3**) (5.00 g, 18.75 mmol) was dissolved in absolute ethanol (40 mL). An excess of 5-methylsalicylaldehyde (3.9 g, 28.7 mmol) was added and the mixture was refluxed for 6 h. After the mixture was cooled to room temperature, the precipitate produced was filtered, washed with water (3 × 30 mL) and dried to give a yellow crude product. The crude product was purified by column chromatography on flash silica gel using dichloromethane–acetone (*v*:*v* = 20:1, *R*_f = 0.73) as the eluent to give 2.1 g (29%) of 4-allylamino-*N*-(*N*-5-methylsalicylidene)-1,8-naphthalimide (**1**); m.p.: 210–211°C. – IR (KBr; *v*/cm^{−1}): 1701, 1653, 1637, 1582, 1534, 1335, 914, 826. – UV/Vis (in DMF, nm): 273, 327, 440. – MS: *m/z* = 386.1857 ([C₂₃H₁₉N₃O₃]⁺1). – ¹H NMR ([D₆]DMSO, 400 MHz): δ (ppm) = 8.91 (s, 1H), 8.73 (d, 1H, *J* = 8 Hz), 8.48 (d, 1H, *J* = 7.2 Hz), 8.28 (d, 1H, *J* = 8 Hz), 7.73–7.75 (m, 1H), 7.57–7.70 (m, 1H), 7.29–7.30 (m, 1H), 6.95 (d, 1H, *J* = 8.4 Hz), 6.75 (d, 1H, *J* = 8.8 Hz), 5.96–6.03 (m, 1H), 5.19–5.31 (m, 2H), 4.09 (s, 2H), 2.29 (s, 3H). – ¹³C NMR ([D₆]DMSO, 400 MHz): δ (ppm) = 169.47, 160.73, 160.11, 156.62, 150.86, 134.70, 134.47, 134.11, 131.23, 131.02, 130.68, 130.51, 128.79, 128.35, 124.56, 124.34, 121.88, 120.10, 117.41, 116.76, 107.50, 44.97, 19.96. – Elemental analysis for C₂₃H₁₉N₃O₃: calcd. C 71.67, H 4.97, N 10.90; found: C 71.63, H 4.98, N 10.93.

4 Conclusions

In conclusion, we have developed a highly selective fluorescence sensor for a Cu²⁺ ion based on the naphthalimide

framework. Compound **1** exhibits a highly selective response to Cu²⁺ over other metal ions in Tris-HCl (pH = 7.2) buffer–DMF (1 : 1, *v/v*) solution with a detection limit of 0.23 μM and an association constant of 1.328 × 10⁶ M^{−1}. UV/Vis titration and fluorescence titration experiments and Job plot analysis of **1** to Cu²⁺ indicate the formation of a 1 : 1 stoichiometric complex. The good selectivity and sensitivity in aqueous medium effectively enhance the application value of the chemosensor for Cu²⁺.

4.1 Supplementary information

Figures of the IR, UV/Vis, NMR and mass spectra of compound **1** are given as Supplementary Information available online (DOI: 10.1515/znb-2016-0138).

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